THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The number of perfect squares between 200 and 900 (both excluded) is: 1

(a) 17 **(b)** 14 **(c)** 15 **(d)** 16

2. 2^7 added 2 times equals: 1

(a) 2^8 **(b)** 4^7 **(c)** 2^9 **(d)** 2^{14}

3. If $(34)_b = 19$ in decimal (a two-digit number in base b), then b equals: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

4. The angles of a quadrilateral are in the ratio 3:4:5:8. The difference between the largest and the smallest angle is: 1

(a) 144° **(b)** 90° **(c)** 102° **(d)** 198°

5. The smallest number which leaves remainder 5 when divided by 12 and 18 is: 1

(a) 77 **(b)** 221 **(c)** 36 **(d)** 41

6. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 81 **(b)** 67 **(c)** 109 **(d)** 53

7. If x : y = 2 : 3 and y : z = 4 : 7, then x : y : z is: 1

(a) 8 : 12 : 21 **(b)** 2 : 3 : 7 **(c)** 2 : 4 : 7 **(d)** 2 : 7 : 7

8. Two articles are sold at ₹ 1800 each, one at a gain of 10% and the other at a loss of 10%. The overall result is a loss of (in %): 1

(a) 0% **(b)** 1% **(c)** 10% **(d)** 5%

9. The legs of a right triangle are in the ratio $5 : 12$ and its hypotenuse is 52 cm. Its perimeter (in cm) is: 1

(a) 124 **(b)** 68 **(c)** 120 **(d)** 104

10. The present ages of Ria and Sia are in the ratio 3:5. After 8 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 22 **(b)** 40 **(c)** 32 **(d)** 24

11. All the 12 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 30 **(b)** 44 **(c)** 20 **(d)** 18

12. The mean of 5 numbers is 30. When one number is removed the mean of the remaining 4 becomes 27. The removed number is: 1

(a) 42 **(b)** 15 **(c)** 30 **(d)** 3

13. A two-digit number exceeds the number formed by reversing its digits by 18. The difference of its digits is: 1

(a) 18 **(b)** 1 **(c)** 3 **(d)** 2

14. The perimeter of a rectangle is 128 cm and the length is 3 times the breadth. Its area is: 1

(a) 736 cm² **(b)** 1024 cm² **(c)** 784 cm² **(d)** 768 cm²

15. The smallest number to be added to 1234 to make it a perfect square is: 1

(a) 64 **(b)** 62 **(c)** 61 **(d)** 63

16. If $2^{2x+1} = 2^9$, the value of x is: 1

(a) 8 **(b)** 4 **(c)** 3 **(d)** 5

17. **Assertion (A):** The number 1296 is a perfect square. 1

Reason (R): A number is a perfect square if every prime in its prime factorisation occurs an even number of times.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

18. **Assertion (A):** $2^{-3} = \frac{1}{8}$. 1

Reason (R): For a non-zero number a, $a^{-n} = \frac{1}{a^n}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. **Assertion (A):** The binary number 101_2 equals 5 in decimal. 1

Reason (R): In base 2, the value of a number is the sum of each digit times a power of 2, with the rightmost place being 2^0.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The adjacent angles of a parallelogram are supplementary. 1

Reason (R): The opposite sides of a parallelogram are parallel, so adjacent angles are co-interior angles on a transversal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

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SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the smallest number to be added to 3900 to make it a perfect square. What is the square root of the result? **2**

22. Find x if $3^{2x-1} = 243$. **2**

23. Find the base b if $(23)_b = 13$ in decimal. **2**

24. In a parallelogram ABCD, $\angle A = (4x + 14)^\circ$ and $\angle C = (5x + 2)^\circ$. Find x and $\angle B$. **2**

25. Find the smallest number which leaves remainder 3 when divided by 20 and 25. **2**

26. If $x + y = 12$ and $xy = 35$, find $x^2 + y^2$. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A square garden has area 1225 m². Fencing costs ₹40 per metre. Find the total cost of fencing it. **3**

28. A sheet of paper 0.1 mm thick is folded in half 10 times (assume this is possible). Find the thickness of the folded paper in mm and in cm. **3**

29. A trader has weights of 1, 2, 4, 8, 16, 32 and 64 g. How can she weigh exactly 45 g? Explain using binary numbers. **3**

30. In a parallelogram ABCD with $\angle A = 50^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. **3**

31. Show that the six-digit number 247247 is divisible by 7, 11 and 13. **3**

OR

If $x - y = 4$ and $xy = 21$, find $x^2 + y^2$ and $(x + y)^2$. **3**

32. The ratio of the present ages of a father and son is 3 : 1. After 4 years the ratio will be 7 : 3. Find their present ages. **3**

OR

A merchant sells two items at ₹ 1980 each, one at a gain of 10% and the other at a loss of 10%. Find the overall gain or loss in rupees. **3**

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Ramesh, a gardener in Jaipur, is given a square plot of area 1764 m^2 to plant a lawn and fence it. He wants to find its dimensions using prime factorisation. **5**

(i) Express 1764 as a product of prime factors.

(ii) Find the side of the plot.

(iii) Find the cost of fencing all four sides at ₹50 per metre.

(iv) The owner decides to increase the side to 45 m. Find the extra area added.

34. **Case study.** The distance of the Earth from the Sun is about $150\,000\,000$ km and light travels at about 3×10^5 km per second. Students of a science club use these to answer questions on standard form. 5

(i) Write $150\,000\,000$ in standard form.

(ii) Find the time light takes to reach the Earth from the Sun.

(iii) Write that time in standard form.

(iv) Light takes about 1.3 s to reach the Earth from the Moon. Find the Moon's distance in standard form.

35. In some base b the number $(234)_b$ equals 69 in decimal. Find b and verify your answer by expanding. 5

36. In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 1 : 2 : 3 : 4$. Find the four angles. Show that AB $\parallel$ DC. 5

OR

Find the digits a and b if the number 4a3b is divisible by 45. List all possibilities. 5

37. The sides of a rectangle are $(3x + 2)$ m and $(3x - 2)$ m. Find its area as an expression, its value at $x = 5$, and by how much it is less than the area of a square of side $3x$ m. 5

OR

A, B and C invest ₹ 2000, ₹ 3000 and ₹ 5000 for 6, 4 and 3 months respectively. A profit of ₹ 1300 is shared in the ratio of (investment $\times$ time). Find the ratio and each share. 5

38. A sum of ₹ 10000 is invested for 3 years at 20% p.a. compounded annually. Find the interest for each of the three years separately and verify that their total is the compound interest. 5

OR

In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 9 cm, BC = 12 cm, CD = 20 cm and DA = 25 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

END OF QUESTION PAPER **Total: 80 Marks**

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